



Review Test Submission: Assignment Week 3

User	Mr W (Wouter) W Nel
Course	EBN 122 S2 2017
Test	Assignment Week 3
Started	8/7/17 8:07 AM
Submitted	8/7/17 12:08 PM
Due Date	8/7/17 11:59 PM
Status	Completed
Attempt Score	20 out of 20 points
Time Elapsed	4 hours, 0 minute

Instructions

Read over the document *ClickUP Assignments Instructions*.

Please direct any enquiries regarding ClickUP assignments to
EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use singular)
- Use 'u' to indicate micro.
- Only use pico(p), nano(n), micro(u), milli(m), kilo(k), mega(M), giga(G) and tera(T); do not use centi, deci and deca

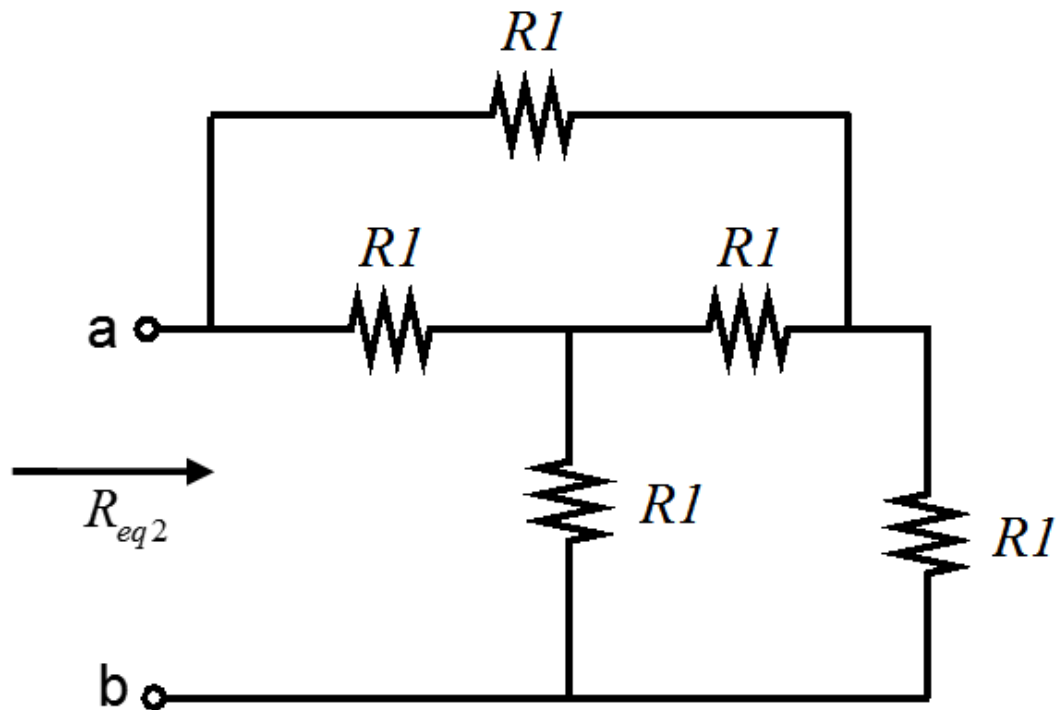
Results Displayed	All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions
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Question 1

2 out of 2 points



Calculate R_{eq2} when $R_I = 82\Omega$.



Selected Answer: ☒ 82ohm

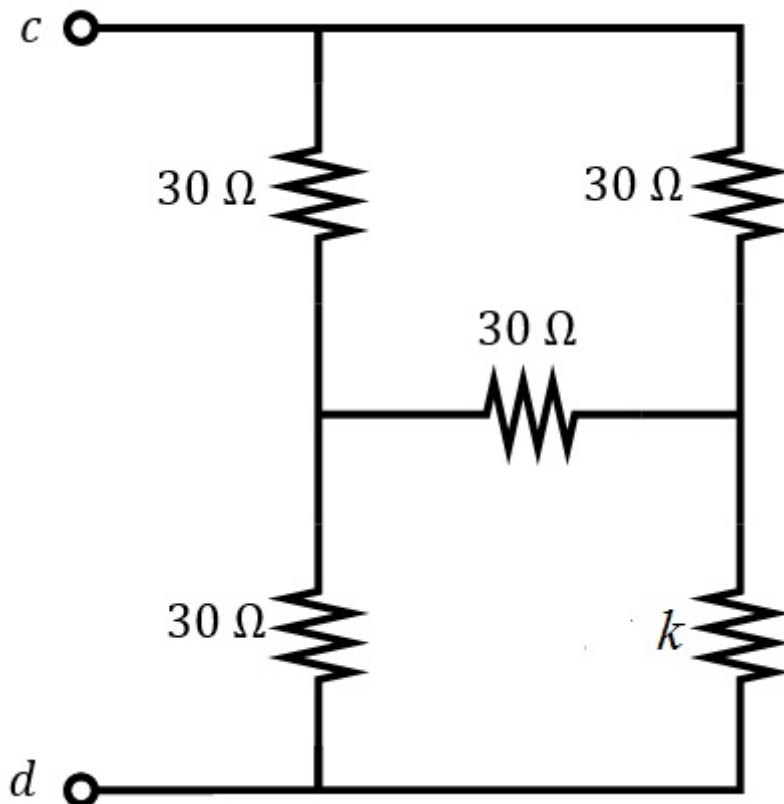
Correct Answer: ☒ 82.00 $\pm$ 2% (ohm)

Question 2

2 out of 2 points



With $k = 1\Omega$ find R_{cd}



Selected Answer: ☒ 18.6ohm

Correct Answer:  $18.63 \pm 5\%$ (ohm)

Question 3

2 out of 2 points



Given $m = 5$, determine V_3 .

$$\begin{bmatrix} 5 & -8 & 1 \\ 2 & 10 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 13 \\ 0 \\ m \end{bmatrix}$$

Selected Answer:  1.25V

Correct Answer:  $1.25 \pm 5\%$

Question 4

2 out of 2 points



If $I = 19$ A, find m .

$$\begin{bmatrix} 5 & 8 \\ -1 & m \end{bmatrix} \begin{bmatrix} I \\ V \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

Selected Answer:  -1.3

Correct Answer:  $-1.30 \pm 5\%$

Question 5

2 out of 2 points



With $k = 18$, calculate V_1

$$\begin{bmatrix} 5 & -3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -1 \\ k \end{bmatrix}$$

Selected Answer:  2.29V

Correct Answer:  $2.29 \pm 5\%$

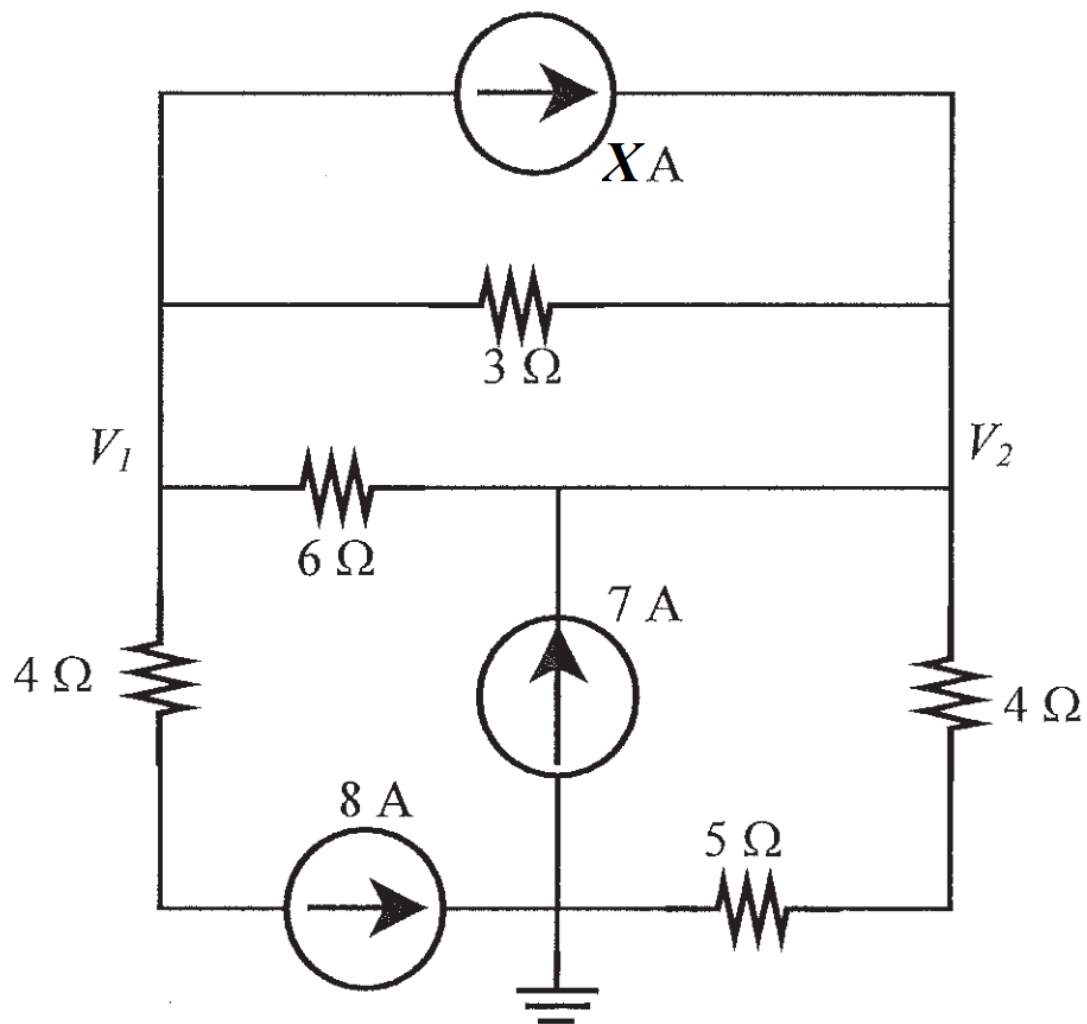
Question 6

2 out of 2 points



Given $X = 9$ A, use nodal analysis to set up an equation of the following form for the circuit. Give the value of b :

$$-9V_1 + 11V_2 = b$$



Selected Answer: ☒ 288

Correct Answer: ☒ $288.00 \pm 5\%$

Question 7

2 out of 2 points

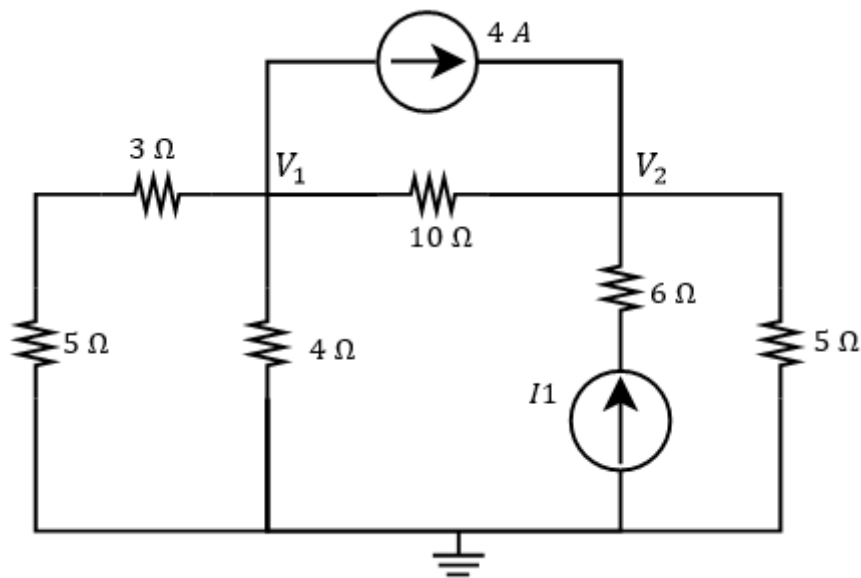


Use nodal analysis to set up two equations of the following form:

☒ Provide e, if $I_1 = 10\text{ A}$.

$$eV_1 + fV_2 = -160$$

$$-V_1 + gV_2 = h$$



Selected Answer: ☒ 19

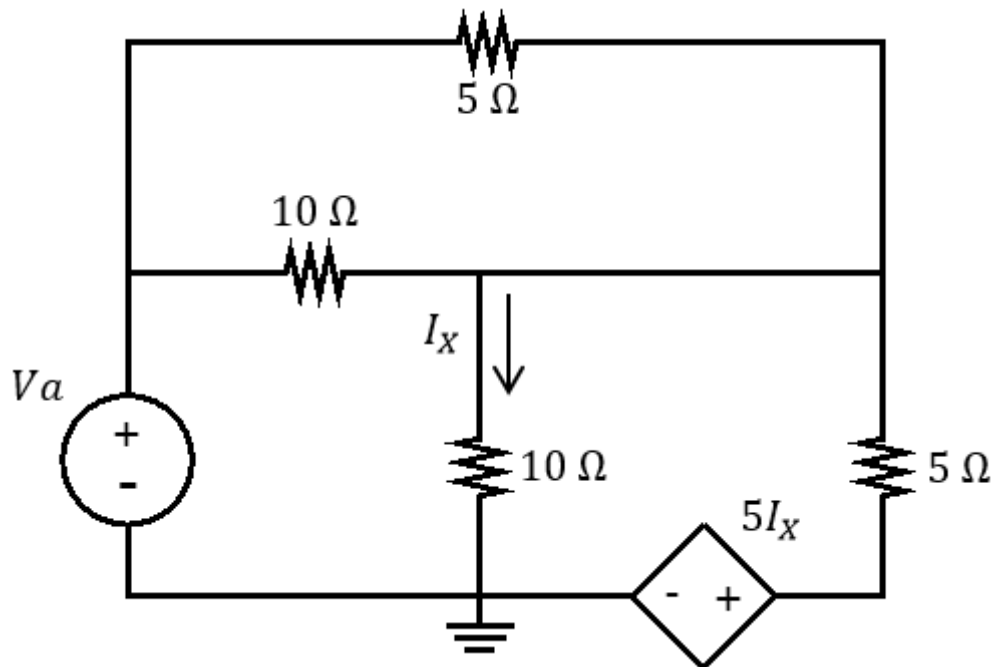
Correct Answer: ☒ 19.00 ± 5%

Question 8

2 out of 2 points



Use nodal analysis to determine I_x , if $V_a = 34$ V.



Selected Answer: ☒ 2.04A

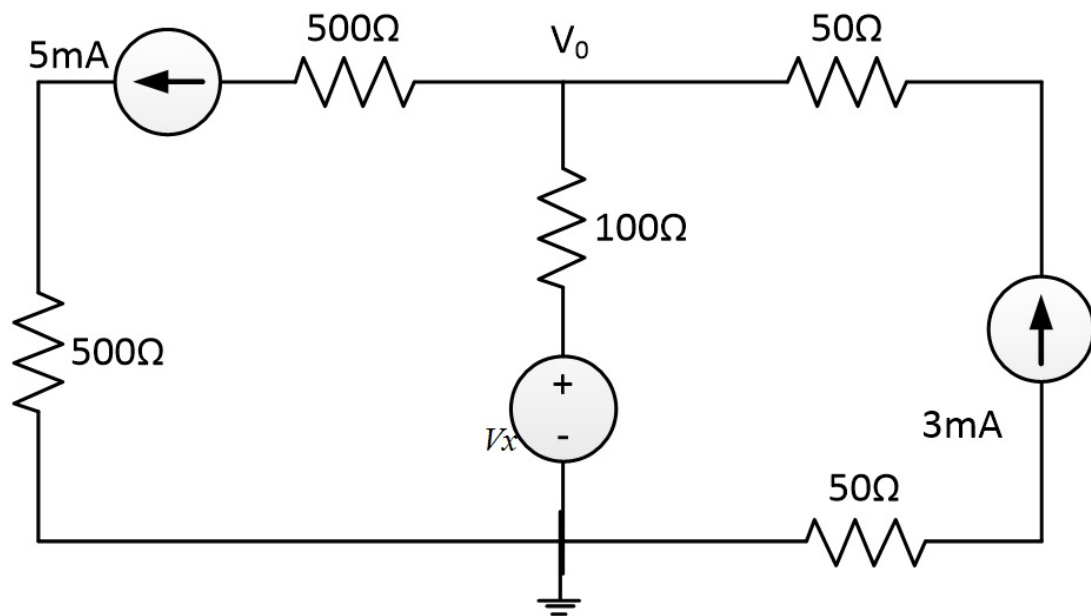
Correct Answer: ☒ 2.04 ± 2% (A)

Question 9

2 out of 2 points



If $V_x = 41$ V find V_o .



Selected Answer: ☒ 40.8V

Correct Answer: ☒ $40.80 \pm 2\%$ (V)

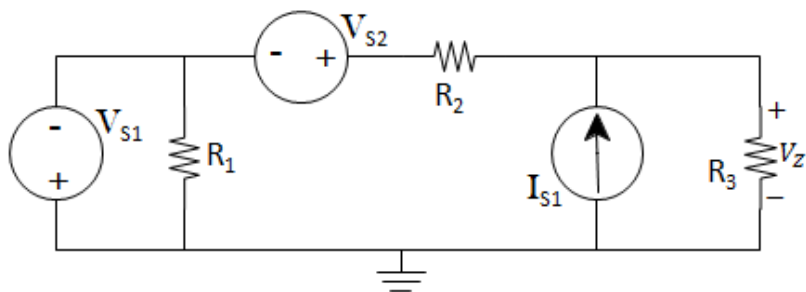
Question 10

2 out of 2 points



☒ Determine the value of V_Z .

$V_{S1}=7V$, $V_{S2}=12V$, $I_{S1}=25mA$, $R_1=15\Omega$, $R_2=11\Omega$, $R_3=64\Omega$



Selected Answer: ☒ 4.5V

Correct Answer: ☒ $4.501 \pm 2\%$ (V)

Monday, August 7, 2017 12:08:45 PM SAST

← OK